

Decision-Level Fusion Based Micro-Motion Target Recognition for One-Bit Radar

Guosen Zhao¹, Bailing Ren², Bin Zhang², Zhaoyang Sun², Hao Liu², Kaidi Song², Bo Zhao^{*1}

¹ The Guangdong Key Laboratory of Intelligent Low-Altitude Perception, The State Key Laboratory of Radio Frequency Heterogeneous Integration, Shenzhen University, Shenzhen 518060, China

² Beijing Research Institute of Telemetry, Beijing 100094, China

Email: zhaoguosen0303@foxmail.com, {bailingren2011, 13581568198, szhyathome, liuhao729}@163.com, 1569956381@qq.com, *b_zhao@126.com

Abstract—In this paper, we propose a real-time high-accuracy one-bit radar micro-motion target recognition method through decision-level fusion. First, a lightweight one-dimensional convolutional neural network (1D-CNN) is designed to extract geometric structure features from the one-dimensional high-resolution range profile (HRRP). Subsequently, a compact micro-convolutional neural network (M-CNN) architecture is developed to capture micro-Doppler characteristics associated with target motion dynamics. To enhance fusion reliability, an adaptive weight allocation mechanism is introduced to refine the basic probability assignment (BPA) function, followed by an improved Dempster-Shafer (DS) evidence theory framework for robust decision-level fusion. Experimental validation using field-acquired radar datasets demonstrates that the proposed method effectively mitigates evidence conflicts in fusion processes while achieving 99.10% recognition accuracy with 40.80 ms processing latency.

Keywords—one-bit radar, micro-motion target recognition, improved Dempster-Shafer

I. INTRODUCTION

With advances in sensing technology, radar systems have evolved from purely military instruments to widely deployed civil sensors. In intelligent transportation and autonomous driving, distinguishing micro-moving objects from static backgrounds has become a central challenge for radar perception [1][2]. This, in turn, creates an urgent demand for high-accuracy, real-time micro-motion target recognition on resource-constrained edge platforms.

Current mainstream radar target recognition techniques mainly comprise micro-Doppler analysis and HRRP approaches. The micro-Doppler effect is primarily determined by target motion characteristics and, due to its low sensitivity to external interference, exhibits strong environmental robustness [1]–[5]. This property makes micro-Doppler signatures a reliable basis for target classification and discrimination [6][9]. However, time–frequency analysis requires long integration times and thus suffers from poor real-time performance. With the improvement of radar resolution, HRRP-based target recognition has attracted considerable attention [10]–[12]. Pan [13] combined CNN with bidirectional stacked recurrent neural networks to classify HRRP of three aircraft types. Liu [14] proposed a hierarchical

temporal feature extraction network to capture temporal dependencies among HRRP range cells. Nevertheless, one-bit HRRP suffer from degraded signal-to-noise ratio, leading to reduced classification accuracy. ZA [15] combined radar micro-motion estimation with scattering-center localization to construct a comprehensive dataset for complex target representation, but did not develop a joint recognition framework that fuses micro-motion features with HRRP.

One-bit quantization has become a research hotspot because it greatly alleviates the data-rate burden of radar systems, and its applications have expanded to multiple radar domains [16]–[18]. Motivated by this, we propose, for one-bit radar signals, a 1D-CNN to learn target geometric structure features from HRRP envelopes. In addition, a M-CNN is designed to extract micro-Doppler motion features from time–frequency representations. An improved DS evidence theory–based decision-level fusion scheme is further developed, in which adaptive weighting coefficients are introduced to modify the BPA functions. This strategy enhances fusion accuracy, enabling high-real-time and high-accuracy micro-motion target recognition with one-bit radar.

II. SIGNAL MODEL

A. Zero-Threshold One-Bit Quantization

One-bit quantization drastically reduces data volume and computational complexity in radar signal processing. In this work, the received echoes are processed using dechirp. The processed echo signal $s_p(t)$ is obtained by mixing the reflected signal $s(t)$ with the reference signal $s_r(t)$:

$$\begin{aligned} s_p(t) &= s(t)s_r^*(t) \\ &= A \text{rect}\left(\frac{t-\tau}{T_p}\right) \text{rect}\left(\frac{t-\tau_r}{T_p}\right) \exp\{j(2\pi f_b t + \psi)\}, \end{aligned} \quad (1)$$

where A is the maximum amplitude of the processed echo signal, $\text{rect}(\cdot)$ represents the rectangular window function, τ is the reflected signal time delay, T_p denotes the pulse width of the signal, τ_r is the reference signal time delay, $f_b = k(\tau_r - \tau)$, $\psi = 2\pi f_0(\tau_r - \tau) + \pi k(\tau^2 - \tau_r^2)$, k is the frequency modulation rate, and f_0 is the carrier frequency.

The one-bit quantization of the echo signal with zero-threshold is performed to obtain the binary echo signal $s_{1\text{bits}}(t)$ as follows [16]:

$$s_{1\text{bits}}(t) = \text{sgn}[\text{Re}(s_p(t))] + j \text{sgn}[\text{Im}(s_p(t))], \quad (2)$$

where $\text{sgn}(\cdot)$ represents the signum function for polarity extraction. Equations (2) can be expressed in complex signal form as follows [17]:

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$$s_{\text{1bits}}(t) = \sum_{n=0}^{\infty} \varepsilon_n A_n (-1)^{\frac{n+3}{2}} \exp \left[j(-1)^{\frac{n+3}{2}} n \nu \right], \quad (3)$$

where n denotes the harmonic order. A_n represents the complex amplitude of the n -th harmonic component, $A_n = \int J_n(\sigma \xi) / \xi d\xi$, $J_n(\cdot)$ is the n -th order Bessel function. And $\varepsilon_0 = 1, \varepsilon_1 = \varepsilon_2 = \dots = 2$. The zero-threshold one-bit quantization nonlinearly preserves signal polarity (-1/+1), distorting time-domain waveforms while generating harmonics but retaining critical frequency-domain information [17].

B. Conventional DS Evidence Theory

The DS evidence theory is a decision-level fusion method primarily employed for uncertainty inference and evidential computation. When an unknown target is identified by a sensor, the inference result is represented as a BPA vector that quantifies the target's potential membership across multiple hypothesis classes. The BPA vector $\mathbf{p}$ as follows:

$$\mathbf{p}(A) = [p(A_1), p(A_2), \dots, p(A_i), \dots], \quad (4)$$

where $p(A_i)$ denotes the probability mass assigned to target class A_i .

The DS evidence theory combines multiple BPAs through decision-level fusion to obtain enhanced target characterization. The theory operates via orthogonal summation, denoted by $\oplus$, on evidentiary BPAs. For a two-sensor system identifying an unknown target A , two BPAs $\mathbf{p}_1$ and $\mathbf{p}_2$ are fused using the DS combination rule:

$$\begin{aligned} p(A_i) &= \mathbf{p}_1(A_i) \oplus \mathbf{p}_2(A_i) \\ &= \frac{\sum_{p_1 \cap p_2 = A_i} p_1(A_i) \cdot p_2(A_i)}{1 - c_f}, \\ c_f &= \sum_{p_1 \cap p_2 = \emptyset} p_1(A_i) \cdot p_2(A_i), \end{aligned} \quad (5)$$

where c_f is the conflict factor that indicates the degree of data conflict between BPAs $\mathbf{p}_1$ and $\mathbf{p}_2$. $c_f \in [0, 1]$, with higher values indicating greater conflict between the two BPAs [22].

III. MICRO-MOTION TARGET RECOGNITION

In this section, a decision-level fusion micro-motion target recognition method using the improved DS evidence theory is proposed. First, the target geometric features in the HRRP envelope and the motion-state characteristics in micro-Doppler signatures are extracted to obtain preliminary recognition results. Subsequently, the improved DS evidence theory is applied for decision-level fusion to derive optimal recognition results.

A. One-Bit HRRP and Micro-Doppler Target Recognition

A range-dimension Fourier transform is applied to the one-bit echo signal to generate a one-bit HRRP. The resulting HRRP data is processed through a moving target indicator filter to isolate higher-threat moving targets. The micro-motion target HRRP is extracted via constant false alarm (CFAR) detection from the clutter-canceled HRRP, followed by stabilization through normalization and alignment to mitigate amplitude and translation sensitivities of the HRRP.

The HRRP signal learns envelope features through a convolutional neural network for target recognition. As

HRRP is one-dimensional, the convolution kernel requires a 1D configuration. This section builds a 1D-CNN architecture comprising convolutional layers, Batch Normalization (BN) layers inserted between convolutional and pooling layers to enhance generalization capability, pooling layers, and fully connected layers, as detailed in Fig. 1.

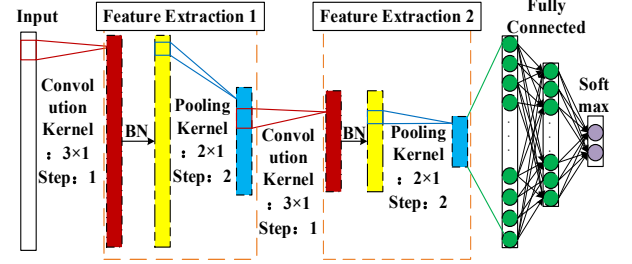


Fig. 1. Architecture of the Lightweight 1D-CNN for HRRP Feature Extraction

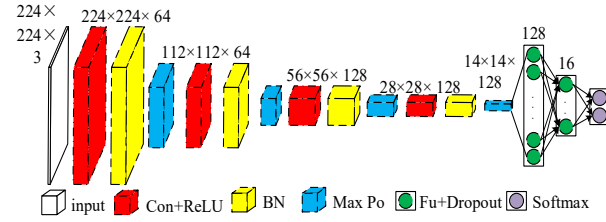


Fig. 2. Architecture of the M-CNN for Micro-Doppler Time-Frequency Feature Learning

Moving target detection is conducted using CFAR for one-bit echo signals suppressing static clutter. The range coordinates of each echo target are captured, with M echo instances recorded to determine the target's range region $[R_a, R_b]$, designated as the micro-motion region. Subsequently, the range-like data of M echoes at $[R_a, R_b]$ are segmented and amplitude-accumulated to derive the time energy vector for this moving target across echo samples. The vector is subjected to time-frequency analysis to extract a representative time-frequency image of the micro-motion.

The standard time-frequency analysis method is the short-time Fourier transform (STFT), the STFT results are squared to compute the modulus, then normalized along the frequency dimension to generate the time-frequency image.

In this section, the VGGNet [23] model is adapted to construct the M-CNN architecture, tailored for processing micro-motion time-frequency images. The modified network employs significantly fewer layers than VGGNet. The complete network architecture is depicted in Fig. 2.

B. Improved DS Evidence Theory Target Recognition

This section implements the improved DS evidence theory for decision-level fusion of HRRP and micro-Doppler recognition results within constrained timeframes, thereby enhancing target recognition precision. The Softmax probabilities directly as singleton BPAs, with no explicit representation of uncertainty.

The conflict factor c_f in DS evidence theory quantifies the degree of inconsistency between two BPAs. When complete conflict occurs between BPAs (i.e., $c_f = 1$), the denominator in (5) becomes zero, rendering the fusion operation undefined.

Analysis reveals that the primary cause of paradoxical

fusion in DS evidence theory stems from zero-probability assignments in BPAs for specific target classes. These null probabilities induce exclusion from conflict factor calculations, failing to fully capture inter-source conflicts. To resolve this paradox, we implement relevance-weighted BPA modifications that adjust zero-probability assignments. The computational workflow of the improved DS evidence theory is detailed in Fig. 3.

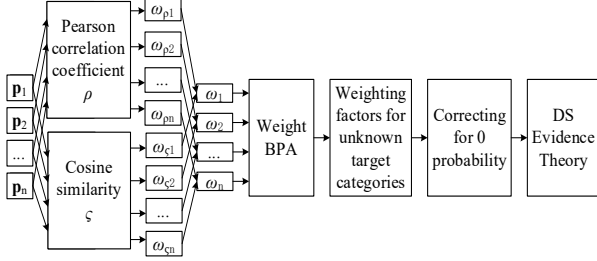


Fig. 3. Relevance-Weighted BPA Modification in Improved DS Evidence Theory for Zero-Probability Correction

The Pearson correlation coefficient (PCC) and cosine similarity (CS) for each BPA are initially computed. To mitigate the influence of negatively correlated BPAs during identification, an adjustment function $\ell(x)$ is formulated to refine the PCC and CS values, defined as:

$$\ell(x) = \begin{cases} x, & x > 0 \\ 0.01, & x \leq 0 \end{cases} \quad (6)$$

After deriving the corrected PCC for each BPA, the PCC matrix ρ is formulated as follows:

$$\rho = \begin{bmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \cdots & \rho_{nn} \end{bmatrix}, \quad (7)$$

where ρ_{ab} denotes the PCC between the a -th $\mathbf{p}$ and b -th $\mathbf{p}$ BPAs. Subsequently, the weighting coefficients for each BPA's PCC ω_{ρ_n} are computed:

$$\omega_{\rho_n} = \frac{\sum_{b=1, b \neq n}^n \rho_{nb}}{\sum_{a=1}^n \sum_{b=1, b \neq n}^n \rho_{ab}}. \quad (8)$$

An analogous procedure is applied to compute the CS matrix and derive the CS weights ω_{ζ_n} for individual BPAs. Leveraging the PCC and CS based weights, the composite weighting ω_{p_n} coefficients are determined through arithmetic averaging:

$$\omega_{p_n} = \frac{\omega_{\rho_n} + \omega_{\zeta_n}}{2}. \quad (9)$$

The derived weights are multiplied by BPAs to compute the weighted averages for each category:

$$\omega_{A_i} = \frac{\sum_{n=1}^n p_n(A_i) \omega_{p_n}}{n}. \quad (10)$$

Subsequently, zero-probability assignments are adjusted using categorical weights. For each BPA, zero-probability categories are identified and probability mass is redistributed from the maximum-probability term proportionally to its weight. For multiple zero-probability terms, reallocation occurs from the maximum-probability term via weighted distribution. When multiple maxima share identical values, redistributed mass is evenly divided. Finally, enhanced fusion

results are generated by applying DS evidence theory to corrected BPA functions.

IV. EXPERIMENTAL

The experimental setup incorporates two micro-motion target types: pedestrians and cyclists. For both target classes, multiple motion patterns were designed, including motion toward the radar along the radar line of sight, motion away from the radar, motion perpendicular to the radar line of sight, and motion at an oblique angle. The bicycles were typical shared bicycles, with an approximate length of 1.5m and a width of about 0.5m. The pedestrians were adult males or females, with heights ranging from 1.6 m to 1.9m and a width of about 0.5m. The velocity of the bicycles ranged approximately from 0.5m/s to 3m/s, whereas the velocity of the pedestrians ranged approximately from 0.5m/s to 2m/s. High-precision and one-bit radar data are synchronously recorded. The data acquisition scenario is shown in Fig. 4.



Fig. 4. The data acquisition scenario: (a) pedestrians (b) cyclists

A. Experiments Results of One-Bit HRRP

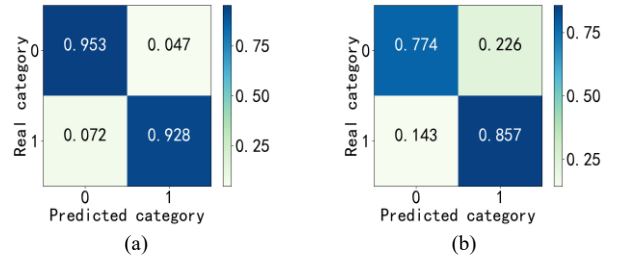


Fig. 5. Test Set HRRP Recognition Confusion Matrices: (a) High-Precision Radar Data (b) One-Bit Radar Data

The 1D-CNN model was employed to train the recorded HRRP data, with the best-performing model preserved. The confusion matrix derived from test set evaluation is presented in Fig. 5. The high-precision radar HRRP test set achieves an average accuracy of 94.05%, demonstrating high efficacy of the 1D-CNN model in extracting discriminative features from micro-motion target HRRP data. In contrast, the one-bit radar HRRP test set yields 81.55% accuracy, representing a 12.50% performance degradation compared to high-precision data. This disparity stems from signal-to-clutter ratio degradation and compromised feature localization induced by one-bit quantization.

TABLE I. TEST SET PERFORMANCE OF ONE-BIT RADAR HRRP CLASSIFICATION WITH FOUR CLASSIFIERS

Classifiers	Cyclist Accuracy	Pedestrian accuracy	Average accuracy
1D-CNN	77.40%	85.70%	81.55%
SVM	79.90%	75.30%	77.60%
1D-VGGNet	88.70%	75.90%	82.30%
RNN	79.40%	80.00%	79.70%

Four classification models are employed for one-bit data recognition. The test set performance metrics across

classifiers are summarized in TABLE I. The 1D-VGGNet architecture comprises 22 layers with 28,806,394 parameters, while the 1D-CNN model utilizes merely 9 layers and 140,610 parameters. Comparative analysis indicates that 1D-CNN achieves superior accuracy with substantially reduced model depth and parameter count.

B. Experiments Results of One-Bit Micro-Doppler

The M-CNN model was employed to process micro-Doppler data with 979.2 ms temporal integration, preserving the optimal model configuration. The confusion matrix derived from test set evaluation is presented in Fig. 6. The recognition results demonstrate that one-bit quantization exerts minimal impact on micro-Doppler characteristics of micro-motion targets, while preserving inherent target motion-state characteristics.

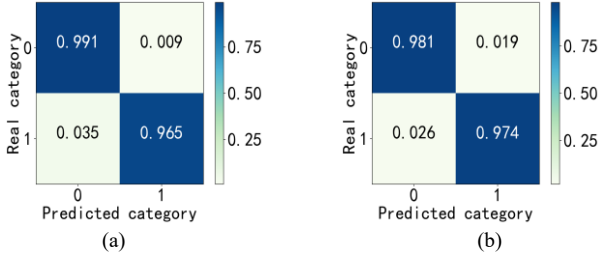


Fig. 6. Test Set Micro-Doppler Recognition Confusion Matrices: (a) High-Precision Radar Data (b) One-Bit Radar Data

Four classification models are employed for one-bit micro-Doppler feature recognition. The test set performance metrics are presented in TABLE II. The VGGNet architecture contains 22 layers with 133,640,954 parameters, while the M-CNN model operates with 15 layers and 263,794 parameters. Analytical results indicate that M-CNN achieves the highest average accuracy while maintaining substantially fewer model layers and parameters.

TABLE II. TEST SET PERFORMANCE OF ONE-BIT RADAR MICRO-DOPPLER CLASSIFICATION WITH FOUR CLASSIFIERS

Classifiers	Cyclist Accuracy	Pedestrian accuracy	Average accuracy
M-CNN	98.10%	97.40%	97.75%
SVM	74.80%	71.10%	72.95%
AlexNet	95.30%	93.90%	94.60%
VGGNet	96.30%	98.20%	97.25%

Although micro-Doppler features of micro-motion targets demonstrate stability and reliability while maintaining satisfactory recognition performance under one-bit quantization, the 979.2 ms temporal integration proves prohibitively excessive for real-time target recognition system. Time-frequency images of pedestrian targets with integration periods of 979.2 ms, 81.6 ms and 4 ms were acquired for systematic comparison, as shown in Fig. 7.

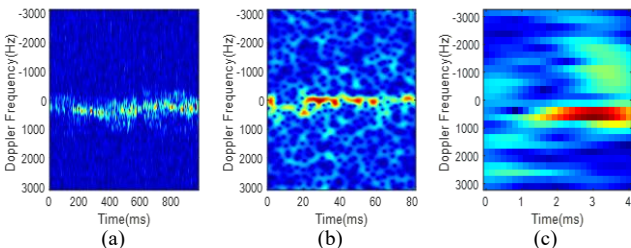


Fig. 7. Time Frequency Images of One-Bit Radar Micro-Doppler Signals with Varied Temporal Integration: (a) 979.2 ms (b) 81.6 ms (c) 4 ms

The M-CNN models are trained on time-frequency images across 12 temporal accumulation durations, with optimal models retained for subsequent test set evaluation. The resultant average accuracy, cyclist accuracy, and pedestrian accuracy curves are presented in Fig. 8. As temporal resolution diminishes, discernible features in time-frequency images inevitably diminish, leading to progressive degradation in target recognition accuracy.

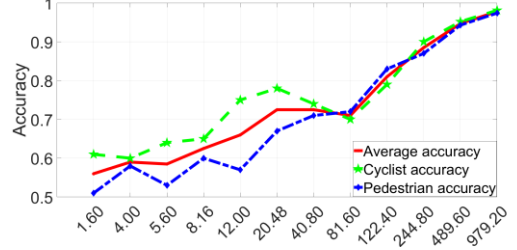


Fig. 8. Accuracy Evolution of One-Bit Micro-Doppler Recognition Across 12 Time Accumulates Intervals

C. Experimental Results of Improved DS Evidence Theory

This section employs the improved DS evidence theory to conduct decision-level fusion of micro-Doppler and HRRP recognition results under low temporal integration, achieving enhanced target recognition accuracy while ensuring real-time operation. To satisfy temporal constraints, seven temporal nodes — 1.60ms, 4.00ms, 5.60ms, 8.16ms, 12.00ms, 20.48ms, and 40.80ms — are analyzed to monitor accuracy evolution post fusion.

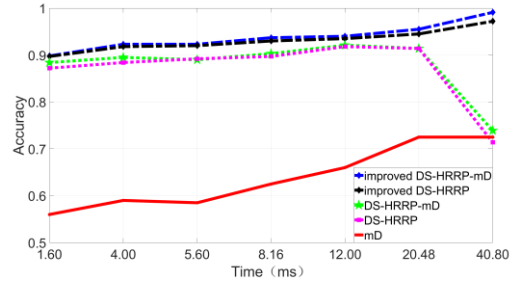


Fig. 9. Accuracy Evolution of Five Multimodal Fusion Strategies Across Seven Temporal Integration Durations

The average accuracy curves for five recognition configurations — improved DS fusion of HRRP and micro-Doppler, conventional DS fusion of HRRP and micro-Doppler, improved DS fusion of HRRP, conventional DS fusion of HRRP, and standalone micro-Doppler recognition — across seven temporal integration intervals (1.60 ms to 40.80 ms) are presented in Fig. 9. Analytical findings reveal that the DS evidence theory experiences evidence conflict at 40.80ms, leading to a decrease in accuracy. The improved DS evidence theory effectively mitigates evidence conflicts, yielding a 99.10% recognition accuracy at 40.80 ms through optimized decision-level fusion.

V. CONCLUSIONS

This paper proposes a frequency-domain feature-based target recognition method for one-bit radar. By employing decision-level fusion, high real-time and high-accuracy recognition of micro-motion targets is achieved with one-bit radar. In the future, extending the dataset to include more diverse target classes is expected to further improve performance and broaden the method's applicability.

REFERENCES

- [1] Patole S M, Torlak M, Wang D, et al, "Automotive radars: A review of signal processing techniques", *IEEE Signal Processing Magazine*, 2017, 34(2): 22-35.
- [2] Wang Y, Jiang Z, Li Y, et al, "RODNet: A real-time radar object detection network cross-supervised by camera-radar fused object 3D localization", *IEEE Journal of Selected Topics in Signal Processing*, 2021, 15(4): 954-967.
- [3] V C Chen, "Analysis of radar micro-Doppler with time-frequency transform," *Proceedings of the Tenth IEEE Workshop on Statistical Signal and Array Processing* . IEEE, 2000: 463-466.
- [4] V C Chen, F Li, S S Ho, H Wechsler, "Analysis of micro-Doppler signatures," *IEE Proceedings-Radar, Sonar and Navigation*, 2003, 150(4): 271-276.
- [5] V C Chen, "Spatial and temporal independent component analysis of micro-Doppler features," *IEEE International Radar Conference*, 2005. IEEE, 2005: 348-353.
- [6] Hanif A, Muaz M, Hasan A, et al, "Micro-Doppler based target recognition with radars: A review", *IEEE Sensors Journal*, 2022, 22(4): 2948-2961.
- [7] M M Johari, M M Nayebi, "Robust airborne target recognition based on recurrence plot quantification of micro-Doppler radar signatures," *2016 17th International Radar Symposium* . IEEE, 2016: 1-4.
- [8] Z Wu and Z. Liu, "RRN-ATR-Net: A Robust Relation Network for Aerial Targets Recognition Based on Two-Stream Micro-Doppler Signatures Fusion," *IEEE Transactions on Instrumentation and Measurement*, 2025, 74: 1-12.
- [9] L. Zhu, S Zhang, K Chen, et al, "Low-SNR Recognition of UAV-to-Ground Targets Based on Micro-Doppler Signatures Using Deep Convolutional Denoising Encoders and Deep Residual Learning," *IEEE Transactions on Geoscience and Remote Sensing*, 2022, 60, 1-13.
- [10] S P Jacobs, "Automatic target recognition using high-resolution radar range-profiles," *Washington University in St. Louis*, 1997.
- [11] L Du, H Liu, Z Bao, J Zhang, "Radar automatic target recognition using complex high-resolution range profiles," *IET Radar, Sonar & Navigation*, 2007, 1(1): 18-26.
- [12] Y Li, H Zhang, W Zhang, et al, "Radar HRRP target recognition based on improved CNN", *Journal of Ordnance Equipment Engineering*, 2022, 43(08): 265-274.
- [13] M Pan, A Liu, Y Yu, et al, "Radar HRRP target recognition model based on a stacked CNN-Bi-RNN with attention mechanism," *IEEE Transactions on Geoscience and Remote Sensing*, 2021, 60: 1-14.
- [14] Q Liu, X Zhang, Y Liu, "Hierarchical sequential feature extraction network for radar target recognition based on HRRP," *2022 7th International Conference on Signal and Image Processing* . IEEE, 2022: 167-171.
- [15] Z A Cammenga, G E Smith, C J Baker, "Combined high range resolution and micro-Doppler analysis of human gait," *2015 IEEE Radar Conference* . IEEE, 2015: 1038-1043.
- [16] J Chen, B Zhao, L Huang, S Liu, W Bao, "Generalization of single-frequency threshold based on 1-bit quantization in synthetic aperture radar," *2021 CIE International Conference on Radar*. IEEE, 2021: 1043-1047.
- [17] B Zhao, L Huang, W Bao, "One-bit SAR imaging based on single-frequency thresholds," *IEEE Transactions on Geoscience and Remote Sensing*, 2019, 57(9): 7017-7032.
- [18] B Zhao, J Chen, L Huang, "One-bit Multi-modality Jamming Method against SAR," *Journal of Radar*, 2022, 11(6): 1119-1130.
- [19] Q Liu, B Zhao, L Huang, "One-bit cross-range scaling approach to ISAR imaging," *IEEE Geoscience and Remote Sensing Letters*, 2024, 21: 1-5.
- [20] W Huang, Q Liu, P Sun, et al, "One-Bit Radar Sparse Imaging Based on Single-Frequency Thresholds," *2024 IEEE 12th Asia-Pacific Conference on Antennas and Propagation (APCAP)*, IEEE, 2024: 1-2.
- [21] G Nie, B Zhao, Q Liu, et al, "One-Bit Synthetic Aperture Radar Imaging Based on Fixed-Threshold With Slow-Time Fluctuations," *IEEE Transactions on Geoscience and Remote Sensing*, 2024, 63: 1-15.
- [22] H Zhang, J G Lu, X H Tang, "An improved DS evidence theory algorithm for conflict evidence," *Journal of Beijing University of Aeronautics and Astronautics*, 2020, 46(3):616-623(in Chinese).
- [23] K Simonyan, A Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, arxiv:1409.1556. [Online] Available: <https://arxiv.org/abs/1409.1556>